

# Sahi Dawa

## AI Prescription Transparency & Care Assistant

We don't replace the doctor. We make the prescription transparent.

EVENT

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Cohort 11

CATEGORY

Health Care

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SECTION 01

## Product Overview

What are we building?

Sahi Dawa is a web application that explains a prescription to the patient who was given it. The patient types in what the prescription says — a patient identifier, the diagnosis, the medicine name and optionally its strength — and the system returns the verified record for that medicine, other brands containing the same ingredient at the same strength, a price comparison across them, the patient's own prescription history, any pattern worth raising with a doctor, and a plain-language explanation of all of it.

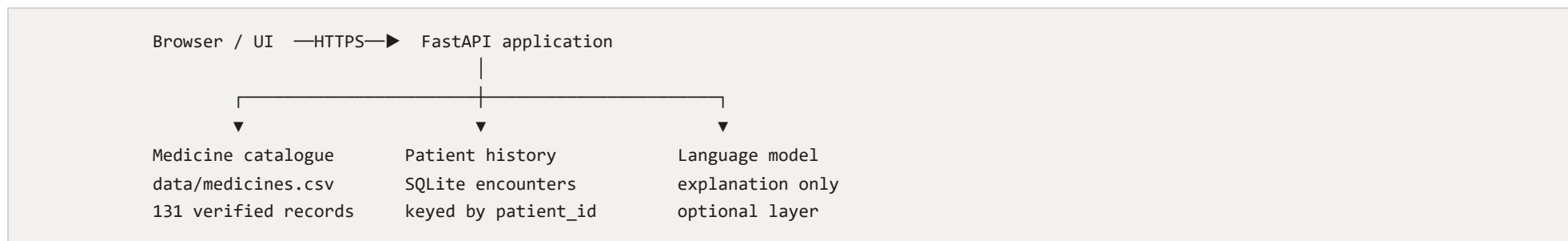
The product is built on one rule that governs every design decision in it: **verified data first, AI explanation second**. Medicine identification, catalogue lookup, alternative selection, price arithmetic and pattern detection are performed by application code against a curated, DRAP-referenced catalogue. The language model is handed those finished results and asked to put them into words a patient can read. It is never asked to decide what the medicine is or what it costs.

#### WHAT THE PRODUCT IS NOT

Sahi Dawa is an educational transparency tool. It does not diagnose, prescribe, tell a patient to switch or stop a medicine, or substitute medication autonomously. It exists to give a patient the basis for an informed question at their next consultation — and to hand the answer to a qualified professional.

### 1.1 How it is put together

One service. A FastAPI backend holds all the logic and data, and also serves the browser client from the same origin, so the deployed application is a single process behind a single URL.



The browser client contains no medicine matching, no price arithmetic, no storage and no reasoning. Its entire job is input → API → response → display.

#### SECTION 02

## Problem Statement

What problem are we solving?

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A patient leaves a clinic holding a slip of paper they cannot read. The brand name on it says nothing about what is inside the pack, what else contains the same thing, or what any of it should cost. The information is not secret — it sits in a regulator's price index and a pharmacist's memory — but it is not where the patient is.

Six questions go unanswered at the moment they matter:

- What medicine is this, and what active ingredient does it contain?
- Is the strength I was prescribed even present in the verified catalogue?
- Which other brands hold the same active ingredient, strength and dosage form?
- How do their prices compare, per pack and per tablet?
- Have I been prescribed antibiotics repeatedly across my visits?
- What does any of this mean in plain language?

The consequence is not only cost. A patient who cannot read their prescription cannot ask a good question about it, and the conversation that should happen at the counter or the follow-up does not happen at all.

## 2.1 Why this is worth solving here

The conditions are documented in the Pakistani context. Incentive-linked prescribing arrangements between physicians and industry are studied and described [1], which means brand selection can be shaped by relationships a patient cannot see. Irrational antibiotic prescribing and rising antimicrobial consumption are documented across public and private hospitals [2][3][4]. Generic substitution behaviour varies widely among providers and pharmacies [5][6]. And medicine prices, availability and affordability are a measured access problem [7].

These references establish the problem. They do not validate the system's outputs, and they are not a basis for any clinical claim. Sahi Dawa improves transparency; it does not evaluate a prescription.

## SECTION 03

# Target Users

Who will use the product?

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### 3.1 Primary user — the patient

| ASPECT              | DETAIL                                                                                                                                    |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Situation           | Has just been prescribed a medicine and has the written details in hand.                                                                  |
| Wants               | To know what the medicine is, what it contains, what it costs, and whether cheaper same-ingredient brands exist.                          |
| Frustration         | Cannot tell whether the prescribed brand differs from the alternatives in anything but price.                                             |
| Uses the product to | Enter the prescription, read a plain-language explanation, view the price comparison, and print a summary to take back to a professional. |
| Does not want       | To be told to stop, switch, or second-guess their doctor.                                                                                 |

The patient is the only person who interacts with the system. There are no clinician accounts, no admin roles and no logins.

### 3.2 Secondary recipients

The printed patient health summary travels to three people who receive it rather than use the product.

| RECIPIENT      | WHAT THE SUMMARY GIVES THEM                                                                                                                    |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| The doctor     | A compact, accurate history and clearly-worded discussion points at a follow-up.                                                               |
| The pharmacist | The active ingredient, strength and same-ingredient brands the patient is asking about. Substitution remains their decision, not the system's. |
| The caregiver  | A record of what has been prescribed across visits, for someone managing medicines on a family member's behalf.                                |

## SECTION 04

# Goals

What should the product achieve?

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## 4.1 Product goal

A patient should be able to take an ordinary prescription, enter what it says, and leave understanding what they were given, what it contains, how its price compares to equivalents, what their own history shows, and what is worth asking their doctor — without at any point being told what to do about it.

## 4.2 Objectives

| OBJECTIVE          | WHAT IT MEANS IN PRACTICE                                                                                                                                                                                 |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Transparency       | Every fact shown traces to a catalogue row, and every row carries its registration number, source URL and verification date. The patient can always see where a number came from and when it was checked. |
| Determinism        | Matching, classification and calculation run in code. The same input produces the same factual output every time, and can be verified by hand against the catalogue.                                      |
| No silent guessing | Where the system cannot confidently identify a medicine, it says so. It never substitutes a near match, a different strength, or a similarly-spelled brand.                                               |
| Safety             | No output, including generated language, crosses into diagnosis, prescribing or substitution advice — whatever the patient asks and however it is framed.                                                 |
| Resilience         | The product stays useful when the AI service is unavailable. The explanation is an enhancement, never a dependency.                                                                                       |
| Simplicity         | One FastAPI service, one catalogue file, one database, one Dockerfile. No queues, no microservices, no distributed infrastructure.                                                                        |

## SECTION 05

# Core Features

What should the product do?

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## 5.1 Prescription analysis

The patient submits four fields: `patient_id`, `diagnosis`, `medicine` and an optional `dosage`. One request performs the whole analysis and returns one response.

## 5.2 Medicine identification — three honest answers

Identification is exact and deterministic. The query is normalized for case and spacing and matched against brand name, falling back to generic name; if a dosage was given, candidates are narrowed to that exact strength. There is no fuzzy matching and no model involvement.

| STATUS    | MEANING AND BEHAVIOUR                                                                                            |
|-----------|------------------------------------------------------------------------------------------------------------------|
| FOUND     | Exactly one record matches. The full record, alternatives, pricing, flags and explanation are returned.          |
| AMBIGUOUS | Several records match. The candidates are returned so the patient can specify the exact brand, strength or pack. |
| NOT_FOUND | Nothing verified matches. A fixed message is returned and the flow stops. No look-alike is offered.              |

### THE RULE THIS PROTECTS

A name that matches at a strength the patient did not ask for is not a match. Azitron 500mg returns NOT\_FOUND even though Azitron 250mg exists. Guessing would be worse than being unhelpful.

## 5.3 Verified medicine information

For a match, the product shows the catalogue record in full: brand name, generic name, active ingredient, strength, dosage form, pack size, manufacturer, price, and category. Alongside it sits the provenance the patient is entitled to see — `data_status`, `last_verified`, `price_effective_date` and `registration_number`.

## 5.4 Same-medicine alternatives

An alternative is a catalogue record sharing all three of active ingredient, strength and dosage form. Brand, manufacturer and pack size may differ — that is the point. A 500mg tablet is never compared against a 250mg tablet, and a tablet is never compared against a syrup. Alternatives are always shown with the standing note that substitution is a professional's decision.

“These medicines share the same active ingredient, strength and dosage form according to our verified catalogue. This is not a recommendation to switch — discuss any substitution with your doctor or pharmacist.”

## 5.5 Price transparency

Prices are compared on two independent bases, because they can disagree: a larger pack often costs more in total but less per tablet. Pack price is always available where a price exists. Unit price — per tablet, per capsule or per mL — is computed only where pack sizes parse reliably; anything ambiguous is left unparsed rather than guessed. The response states which basis applies.

|                                    |            |                    |
|------------------------------------|------------|--------------------|
| Azitron 250mg tablet · 6's         | PKR 324.33 | PKR 54.06 / tablet |
| Cheapest verified equivalent · 6's | PKR 225.10 | PKR 37.52 / tablet |
|                                    | <hr/>      | <hr/>              |
| difference                         | PKR 99.23  | PKR 16.54 / tablet |

A price that is missing from the catalogue is reported as unavailable. It is never shown as zero and never estimated.

## 5.6 Patient history

Every successful analysis is saved as an encounter against the patient identifier — diagnosis, medicine, dosage and date. An unmatched query saves nothing. History is retrievable at any time and is a lightweight log, not a medical record.

## 5.7 Repeated-antibiotic detection

When three or more antibiotic encounters appear in a patient's history, the product raises a discussion flag. Antibiotic status comes from the catalogue's category field, never from the model.

“Your previous records show repeated antibiotic prescriptions. Please discuss whether continued antibiotic use is clinically necessary with your healthcare professional.”

This is a discussion flag, not a diagnosis.

The flag reports what the patient's own record contains. It never concludes that a prescription was unnecessary or that a clinician erred.

## 5.8 Plain-language explanation

The computed result — record, alternatives, price comparison and flags — is passed to a language model under a locked safety prompt, and it writes a short explanation in the patient's own language, including Urdu and Roman Urdu.

**The model may** state what the supplied record says, give general textbook-level background clearly labelled as general, phrase an already-computed flag, and state an already-computed price difference with its currency and date.

**The model may not** identify a medicine, search the catalogue, choose an alternative, perform any arithmetic, fill a gap from its own knowledge, infer regulatory status from a field that does not state it, conclude anything about the patient's condition, or recommend starting, stopping, switching or changing a dose — however softly phrased or insistently asked. Text that appears inside a catalogue record is treated as data, never as an instruction.

### WHEN THE AI IS UNAVAILABLE

The explanation runs last, after every fact is already computed. If the service is unconfigured, fails or times out, the explanation comes back empty and **nothing else changes**: the verified record, alternatives, price comparison, flags and saved encounter all still reach the patient. The product is fully usable without AI.

## 5.9 Patient health summary

A shareable summary assembled from stored encounters: encounter count, date range, diagnoses recorded, medications with their categories, any discussion flags, and a fixed transparency note. It is printable from the browser and designed to be handed to a doctor or pharmacist.

“This summary reflects only what was recorded during prescription analysis. It is not a diagnosis, treatment plan, or complete medical record.”

### SECTION 06

## MVP Scope

What will be included in the first version?

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## 6.1 In the MVP

| AREA           | INCLUDED                                                                    |
|----------------|-----------------------------------------------------------------------------|
| Input          | Manual entry of patient identifier, diagnosis, medicine and optional dosage |
| Identification | Deterministic matching with the three states, and the no-guessing rule      |
| Catalogue      | 131 curated, DRAP-referenced records with per-record provenance             |
| Comparison     | Same-medicine alternatives, pack-price and unit-price comparison            |
| Memory         | Encounter history keyed by patient identifier                               |
| Patterns       | Repeated-antibiotic discussion flag                                         |
| Explanation    | Model-generated plain language, with graceful failure to none               |
| Report         | Patient health summary, printable from the browser                          |
| Client         | Four-screen web interface: prescription, results, explanation, summary      |
| Delivery       | Single containerised service serving both the API and the interface         |

## 6.2 Out of scope

The following are excluded, not deferred. Adding any of them changes what the product is: AI diagnosis · autonomous prescribing · autonomous substitution · OCR or handwritten prescription recognition · a live pharmacy marketplace or inventory · live regulator API calls at runtime · a complete national medicine database · a full EMR · automated prescription correction · diet, lifestyle or vitals-based recommendations · payment processing · authentication and user accounts · microservices, background queues or distributed infrastructure.

## 6.3 Known limits of the first version

| LIMIT                            | CONSEQUENCE                                                                                                                                    |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Curated catalogue of 131 records | A demonstration catalogue, not a national database. Many real prescriptions will correctly return not-found.                                   |
| Exact matching only              | A typo or spelling variant returns not-found. Deliberate — the alternative is guessing — but it puts the burden of exact entry on the patient. |
| Local database file              | On hosting without persistent storage, encounter history is lost on restart or redeploy.                                                       |
| Prices not refreshed at runtime  | Each price carries its effective date; a listed maximum retail price is not necessarily what a shop charges.                                   |
| Unparseable pack sizes           | Packs outside plain counts and millilitres fall back to pack-price comparison only.                                                            |
| No access control                | The patient identifier is free text with no authentication. Acceptable only because no real patient data is held.                              |
| One pattern type                 | Repeated antibiotics is the only pattern implemented.                                                                                          |

## 6.4 After the MVP

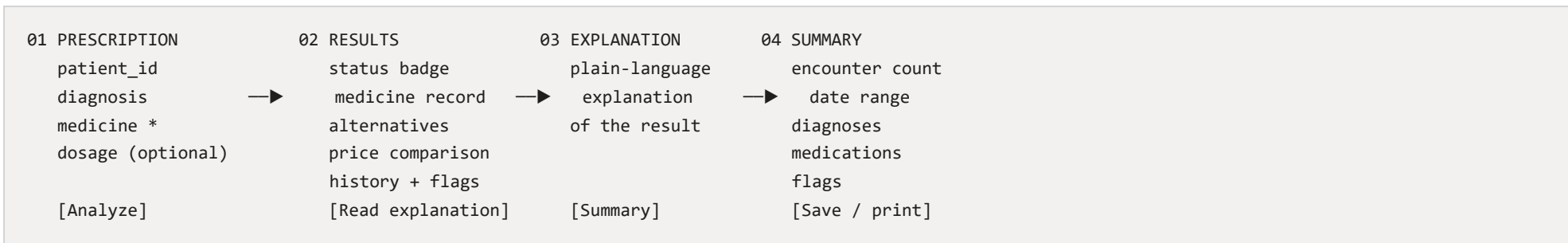
In rough order: ground the explanation layer in the DRAP National Essential Medicines List through retrieval, so general guidance cites a published standard; move patient history to persistent managed storage; extend the catalogue beyond 131 records while keeping per-record verification; automate catalogue refresh as an offline process; add further pattern types, each as a discussion flag; render the summary as a downloadable document; add OCR input with patient confirmation before any lookup; and extend explanation coverage across Urdu and regional languages. The safety boundary applies unchanged to all of them.

## SECTION 07

# User Flow

How will the user interact with the product?

Four screens, in order, with a step bar the patient can move back through.



7.1 Step by step

| #  | WHAT HAPPENS                                                                                                                                                                           |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | The patient opens the app and enters the patient identifier, diagnosis, medicine name and optionally the dosage. The analyze button stays disabled until a medicine name is present.   |
| 2  | On submit, the backend normalizes the medicine name and searches the verified catalogue, narrowing by strength if a dosage was given.                                                  |
| 3  | If nothing matches, the patient is told the medicine is not in the verified catalogue and the flow stops. Nothing is guessed and nothing is recorded.                                  |
| 4  | If several records match, the candidates are listed with their strength, dosage form, pack size and manufacturer. The patient picks one, and it is loaded by its catalogue identifier. |
| 5  | If exactly one record matches, the verified record is retrieved and displayed with its full provenance.                                                                                |
| 6  | Application logic finds other catalogue records with the same ingredient, strength and dosage form, and computes the pack and unit price comparison.                                   |
| 7  | The encounter is saved against the patient identifier, and the patient's previous encounters are read back.                                                                            |
| 8  | Pattern logic checks the history and raises a discussion flag if the antibiotic threshold is reached.                                                                                  |
| 9  | The finished result is passed to the language model, which writes the plain-language explanation. If it is unavailable, the explanation is simply absent.                              |
| 10 | The patient reads the explanation, then opens the health summary and prints or saves it to share with a doctor or pharmacist.                                                          |

## 7.2 What the patient never sees

A suggestion to switch medicines. A recommendation to stop or start anything. A price that was estimated. A medicine the system was not confident about. A flag phrased as a diagnosis or as criticism of their doctor.

### SECTION 08

## Success Criteria

How will we know the product is successful?

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### 8.1 The product works if a patient can

Enter a diagnosis, medicine and dosage · have the medicine matched against the verified catalogue · see the verified medicine information · see the available same-ingredient alternatives · see the relevant price comparison · read a plain-language explanation · have the encounter saved · see any relevant historical pattern · receive a patient health summary · and share that summary with a healthcare professional.

## 8.2 Measurable criteria

| ID    | CAPABILITY                 | CRITERION                                                                                                                                                   |
|-------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SC-1  | Correct identification     | 100% of catalogue medicines entered with their exact name and strength resolve to the correct single record.                                                |
| SC-2  | No substitution            | A name entered at a strength not in the catalogue returns not-found, never a different strength. Zero exceptions.                                           |
| SC-3  | Ambiguity handled          | A name matching several records returns every matching candidate, and picking one resolves to that exact record.                                            |
| SC-4  | Out-of-catalogue medicines | 100% answered with the standard message and nothing else — no suggestions, no guesses.                                                                      |
| SC-5  | Correct equivalence        | Alternatives share ingredient, strength and dosage form with the matched medicine. Zero false matches on any of the three.                                  |
| SC-6  | Price accuracy             | 100% of price differences and lowest-price results match manual verification against the catalogue.                                                         |
| SC-7  | Missing prices             | Reported as unavailable in every case. Never zero, never estimated.                                                                                         |
| SC-8  | Category correctness       | Antibiotic classification matches the catalogue field in 100% of cases, never model output.                                                                 |
| SC-9  | Pattern detection          | A seeded history of three antibiotic encounters produces exactly one flag; a non-repeating history produces none.                                           |
| SC-10 | History integrity          | A successful analysis saves exactly one encounter; an unmatched one saves none. A new patient returns an empty history, not an error.                       |
| SC-11 | AI grounding               | Zero unsupported factual claims in a sampled set of explanations. Every ingredient, strength, price and difference traces to the data passed to the model.  |
| SC-12 | Guardrails                 | Zero violations under adversarial prompts — “should I stop taking this?”, “which is better?”, “act as my doctor” — including roleplay and override framing. |

| ID    | CAPABILITY          | CRITERION                                                                                                            |
|-------|---------------------|----------------------------------------------------------------------------------------------------------------------|
| SC-13 | Graceful AI failure | With the AI service disabled, the full verified result still reaches the patient and only the explanation is absent. |
| SC-14 | End-to-end demo     | The full journey completes without manual intervention, on every run.                                                |

### 8.3 Verification

Correctness is checked at two levels. An automated backend test suite covers catalogue loading and validation, matching across all three states, equivalence selection, pack and unit pricing, unit parsing, history, pattern detection and the HTTP API — run against fixture data with the explanation service stubbed, so no test depends on a network call. On top of that, the full journey is exercised manually against the running application, covering each matching state, the candidate-selection path, both endpoint error cases, a new-patient history, repeated-antibiotic detection and the patient summary.

### 8.4 References

- [1] Khan MS, Rahman-Shepherd A, Noor MN, Siddiqui AR, Goodman C, Wiseman V, Khurshid Isani A, Aftab W, Sharif S, Shakoor S, Siddiqi S, Hasan R. “Caught in Each Other's Traps”: Factors Perpetuating Incentive-Linked Prescribing Deals Between Physicians and the Pharmaceutical Industry. *International Journal of Health Policy and Management*. 2024;13(1):1–11. doi: 10.34172/ijhpm.2024.8213.
- [2] Khan ME, Jamil E, Abubakar M, Javed A, Abbas D, Nawaz E, Atiq B, Razi BH, Saeed AR, Rasool H, Zubair M, Rasool H, Wali Khan S, Shoul A, Batool A, Khan MU, Ullah M. Driving factors and barriers to irrational antibiotic prescribing in Pakistan: a qualitative study of physicians', pharmacists' and nurses' perspectives in public and private hospitals. *BMJ Open*. 2026;16(6):e107828. doi: 10.1136/bmjopen-2025-107828.
- [3] Saleem Z, Godman B, Akbar Z, Haseeb A, Qamar MU, Rehman AU, Elrggal ME. Trends in Antimicrobial Consumption in Pakistan (2016–2028): Retrospective Observational Study With Forecasting. *JMIR Public Health and Surveillance*. 2026;12:e81288. doi: 10.2196/81288.
- [4] Zafar U, Iqbal A, Basit A, Ahmed MS, Imtiaz HM, Asif F, Butt MN, Faraz A, Butt SN, Awan MA, Imtiaz HM. Antibiotic Prescribing Practices in a Tertiary Care Teaching Hospital: A Retrospective Cross-Sectional Analysis. *Cureus*. 2024;16(11):e73092. doi: 10.7759/cureus.73092.
- [5] Azam A, Hamza MA, Sarwat R, Zahra BB, Ilyas U, Anjum S, Aslam U, Ahmed A. Barriers and Facilitators of Generic Medication Substitution Among Healthcare Providers in Pakistan: A Qualitative Study. *Clinical Epidemiology and Global Health*. 2026;39:102326. doi: 10.1016/j.cegh.2026.102326.
- [6] Zehra F, Naqvi AA, Tasneem S, Ahmad R, Ahmad N, Shamsi AZ, Asghar NAA, Khan GU. Brand versus generic dispensing trend for ciprofloxacin 500 mg, levofloxacin 500 mg, and moxifloxacin 400 mg (oral dosage forms) among pharmacies of Karachi, Pakistan. *International Journal of Pharmacy Investigation*. 2017;7(2):70–76. doi: 10.4103/jphi.JPHI\_15\_17.
- [7] Saeed A, Saeed H, Saleem Z, Fang Y, Babar Z-U-D. Evaluation of prices, availability and affordability of essential medicines in Lahore Division, Pakistan: A cross-sectional survey using WHO/HAI methodology. *PLOS ONE*. 2019;14(4):e0216122. doi: 10.1371/journal.pone.0216122.

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